Indian Institute of Technology Jodhpur

No. IITJ/OA/C/2021-22/31/499 **Date: 15.03.2022**

CIRCULAR

Subject: 4-year B.Tech. in X + Engineering Science for B.Tech. students joined post July 2020

I. 4-year B.Tech. in X + Engineering Science

Students can pursue a capability linked B.Tech. in X + Engineering Science to pursue newer structures of programmes in addition to classical programmes, where 'X' represents the primary branch in which a student has taken admissions. After completion of the first year, the student having a CGPA of 6.5 or more may opt to choose Capability-linked B.Tech. in X + Engineering Science. Students will be required to register for extra credits for the same. The students pursuing this programme will take common courses and courses with a concentration in Engineering Science in the 2nd year. From the third year onwards the student would also start opting for courses related to the primary branch (X) and elective courses in the area of both primary branch and Engineering Science. At the end of the second year, an option of primary branch change will also be available to meritorious students of this programme. The students will be awarded B.Tech. in X and Engineering Science, e.g., B.Tech. in Mechanical Engineering and Engineering Science. Students of this programme would not be able to opt for any other specialization/minor.

Course Categories and credit distribution of the proposed Engineering Science Programmes are given in Table 1.

Table 1. Course Categories and credit distribution of X+ Engineering Science Programmes

	Course Type	Course Category	B.Tech. Regular		B.Tech. in X + ES	
			Cr	To	Cr	To
1	Institute Core (I)	Engineering (IE)	34	**69**	34	**59**
		Science (IS)	16		16	
		Humanities (IH)	12		9	
2	Programme Linked (L)	Science (LS)	7		0	
3	Programme Core (P)	Programme Compulsory (PC)	50-52	**71**	50-52	**71**
		Programme Electives (PE)	18-16		18-16	
		B.Tech. Project (PP)	3		3	
4	Open (O)	Open Electives (OE)	10	**10**	-	**-**
5	Engineering Science (E)	Engineering Science Core (EC)	0	**0**	22	**30**
		Engineering Science Elective (EE)	0		8	
		Total Graded		**150**		**160**
6	Non-Graded (N)	Humanities (NH)	6	**15**	6	**15**
		Engineering (NE)	3		3	
		Design/Practical Experience (ND)	6		6	
		Total Graded + Non-Graded		**165**		**175**

Note: Data Structures and algorithms (4 credits) being a common core course for both X and Engineering science, It will be double-counted towards the core course requirement for the award of the degree. In lieu of the same students will be required to take an extra 4 credits worth of engineering science electives.

II. Branch Change for B.Tech. in X +Engineering Science post 2nd year

1. An undergraduate student opted for B.Tech. in X +Engineering Science post 2nd year is eligible to apply for branch change at the end of the fourth semester only, provided the student satisfies the following criteria:
2. Secured a CGPA, at the end of the fourth semester, of:
    1. Greater than or equal to 7.0 for SC/ST and PD applicants
    2. Greater than or equal to 8.0 for other applicants
2. Completed all prescribed course credits for the first four semesters including the non-graded courses (exception, X grade in non-graded courses).
3. No backlog at the end of the fourth semester.
4. The student should have no disciplinary action against him/her.
5. Change of the branch will be permitted strictly in the order of merit as determined by CGPA at the end of the fourth semester. Each available seat may be occupied by students of any category. For students of the SC/ST/PD category, concession in eligibility criteria is given as per rule (a).
6. In case more than one student applying for branch change have the same CGPA, the tie shall be resolved on the basis of JEE ranks of such applicants.
7. In making a change of branch, the strength of a branch should not fall below **95%** of the present strength and should not go above **105%** of the present strength.
8. The conditions mentioned in item (a) above will not be insisted upon for change to a branch in which a vacancy exists with reference to the sanctioned strengths, and the concerned student was eligible as per JEE Rank for admission to that branch at the time of entry to IIT Jodhpur.
9. Notwithstanding any of the conditions listed above, a student with a CGPA of **9** or more at the end of the fourth semester will be allowed to move to a branch of his/her choice.

III. Engineering Science Compulsory and Elective Courses

- **Engineering Science compulsory courses**
    1. Probability, Statistics and Stochastic Processes (3-1-0)
    2. Modern Physics (3-0-0)
    3. Embedded Systems and IoT (3-0-2)
    4. Data Structures and Algorithms (3-0-2)
    5. Design and Analysis of Experiments (3-0-0)
    6. Modelling and Simulation (3-0-2)
- **Engineering Science Elective courses**

    Courses with Science Concentration
    1. Scientific Computations
    2. Foundations of Quantum Information Processing
    3. Physics of Complex Liquid
    4. Analytical Mechanics
    5. Nanoscience
    6. Electronic Materials
    7. Magnetic Materials
    8. Energy Materials
    9. Optical Measurement Techniques

    Courses with Interdisciplinary Concentration
    10. Principles of Biomechanics
    11. MEMS and Microsystems
    12. Non-Conventional Sources of Energy and Emerging Technologies
    13. Modern Control
    14. Sensors and Actuators
    15. Measurement and Instrumentation
    16. Digital Image processing
    17. Computer Vision
    18. Autonomous Systems
    19. Introduction to Industry 4.0
    20. Foundations of Cyber-Physical Systems
    21. Smart Manufacturing
    22. Introduction to Robotics
    23. Introduction to Data Science

Note: In the case of a course being the common core of the primary branch and Engineering science, the course would be double-counted towards the core requirement of both. However, students will be required to take extra credit equivalent to total credits of all courses with double counting under the Engineering Science elective category. For example, students of Mechanical engineering have Data Structures and algorithms (4 credits) as a common core course for both mechanical engineering and Engineering science. Hence, this course will be double-counted towards the core course requirement for the award of the degree. In this case, students will be required to take an extra 4 credits worth of engineering science electives.

IV. Programme Structure of 1st and 2nd year of B.Tech. in X + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester						**II Semester**				
As per the curriculum of Primary Branch X [33 Graded Credit and 6 Non-graded Credits]											
	III Semester						**IV Semester**				
IE	Computational Material Design	1-0-0	1	-	1	IE	Materials Selection	1-0-0	1	-	1
EC	Probability, Statistics and Stochastic Processes	3-1-0	4	-	4	IE	Structure of Materials	1-0-0	1	-	1
EC	Modern Physics	3-0-0	3	-	3	EC	Embedded Systems and IoT	3-0-2	5	-	4
IE	Thermodynamics	3-1-0	4	-	4	EC	Data Structures and Algorithms	3-0-2	5	-	4
IE	Signals and Systems	3-1-0	4	-	4	EC	Design and analysis of Experiments	3-0-0	3	-	3
IE	Introduction to Machine Learning	3-0-2	5	-	4	EC	Modelling and Simulation	3-0-2	5	-	4
NE	Intro. To Profession	0-0-2	2	1		IH	Humanities I	3-0-0	3	-	3
	Total	**16-3-4**	**23**	**1**	**20**		**Total**	**17-0-6**	**23**	**-**	**20**
										Total of graded and Non-Graded Credit	**7**	**73**
										Non-Graded Design Credits	**7**	**-**
										Grand Total	**7**	**73**

V. Programme Structure of 3rd and 4th year of B.Tech. in X + Engineering Science

1) B.Tech. in AI and DS + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						**VI Semester**				
PC	Maths for Computing	3-1-0		-	4	PC	Deep Learning	3-0-3		-	4.5
PC	Design and Analysis of Algorithms	3-1-0			4	PC	Human Machine Interaction Lab	0-0-4			2
PC	Artificial Intelligence	3-0-0			3	PC	PCS-1	2-0-2			3
PC	Optimization	3-0-3			4.5	PC	PCS-2	3-0-2			4
PC	Data Engineering	3-0-3			4.5	PC	Dependable AI	3-0-0			3
						PC	AI+X / DS+X	3-0-0			3
IH	Humanities II	3-0-0	3	-	3	PE/ EE	Programme /Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			**1**	**23**		**Total**			**1**	**22.5**
	VII Semester						**VIII Semester**				
PC	AI+X / DS+X	3-0-0		-	3	PC	Data Visualization	3-0-3		-	4.5
PP	B. Tech. Project	0-0-3	3	-	3	PE/EE	Programme /Engineering Science Electives			-	12
PE/EE	Programme /Engineering Science Electives			-	14						
IS	Environmental Science	2-0-0	2	-	2	IH	Humanities III	3-0-0	3	-	3
	Total	**15-0-6**		**-**	**22**		**Total**	**20-0-0**	**20**	**-**	**19.5**
										Total of graded and Non-Graded Credit	**9**	**160**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**15**	**175**

2) B.Tech. in Bioengineering + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						**VI Semester**				
PC	Biochemistry	3-0-0	3	-	3	PC	Deep Learning	3-0-3	6	-	4.5
PC	Microbiology	3-0-2	5	-	4	PC	Biophysics & Structural Biology	3-0-1	4	-	3.5
PC	Concepts & Dynamics: Molecular Cell Biology	2-0-2	4	-	3	PC	Introductory Omics	3-0-0	3	-	3
PC	The Human Machine for Engineers: Quantitative Physiology	3-0-0	3	-	3	PC	Biomaterials Engineering	3-0-0	3	-	3
PC	Computational and System Biology	3-0-2	5	-	4	PC	Biochemistry Laboratory	0-0-2	2	-	1
PE/ EE	Programme/Engineering Science Electives	3-0-0		-	3	PE	Genetics and Gene Manipulation	3-0-2	5	-	4
IH	Humanities II	3-0-0	3	-	3		The Human Immune System: Mechanisms to Detect, Defend and Attack	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			**1**	**23**		**Total**	**15-2-8**		**1**	**22**
	VII Semester						**VIII Semester**				
PC	Biosensors	3-0-2	5	-	4	PC	Bioimaging	3-0-0		-	3
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives			-	14
PE/ EE	Programme/Engineering Science Electives			-	13	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total	**15-0-6**		**-**	**22**		**Total**	**20-0-0**	**20**	**-**	**20**
										Total of graded and Non-Graded Credit	**9**	**160**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**15**	**175**

3) B.Tech. in Chemical Engineering + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						**VI Semester**				
PC	Mass Transfer I	3-0-3	6	-	4.5	PC	Mass Transfer II	3-0-3	6	-	4.5
PC	Heat Transfer	3-0-0	3	-	3	PC	Process Plant Design & Economics	3-0-3	6	-	4.5
PC	Chemical Reaction Engineering	3-0-3	6	-	4.5	PC	Transport Phenomena	3-1-0	4	-	4
PC	Fluid Mechanics	3-1-0	4	-	4	PC	Process Control and AI Applications	3-0-2	5	-	4
PC	Chemical Engineering Thermodynamics	3-0-0	3	-	3	PC	Fluid Mechanics Lab	0-0-2	2	-	1
PC	Heat Transfer lab	0-0-2	2	-	1	PC	Material & Energy Balance	3-0-0	3	-	3
IH	Humanities II	3-0-0		-	3	PE/ EE	Programme /Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total	**18-2-8**	**27**	**1**	**23**		**Total**	**18-2-10**	**29**	**1**	**24**
	VII Semester						**VIII Semester**				
PC	Scientific Computations	3-0-2	5	-	4	PC	Molecular Engineering	3-0-0	3	-	3
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives			-	14
PE/ EE	Programme/Engineering Science Electives			-	11	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0		-	2						
	Total			**-**	**20**		**Total**		**20**	**-**	**20**
										Total of graded and Non-Graded Credit	**9**	**160**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**15**	**175**

4) B.Tech. in Civil and Infrastructure Engineering + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						**VI Semester**				
PC	Mechanics of solids	3-1-0	4	-	4	PC	Water Resources Engineering	3-0-2	5	-	4
	Transportation Engineering	3-0-0	3	-	3						
	Structural Analysis	3-0-2	5	-	4		Reinforced Concrete Design	3-0-2	5	-	4
	GeoTechnical Engineering	3-1-0	4	-	4		Fluid Mechanics	3-1-0	4	-	4
							Fluid Mechanics Lab	0-0-2	2	-	1
							Advanced Structural Analysis	3-0-2	5	-	4
	Geoinformatics	3-0-2	5	-	4		GeoTechnical Engineering Lab	0-0-2	2	-	1
IH	Humanities II	3-0-0	3	-	3	PE/ EE	Programme/Engineering Science Electives			-	6
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total	**18-3-4**	**24**	**1**	**22**		**Total**			**1**	**24**
	VII Semester						**VIII Semester**				
PC	Construction Technology and Practices	3-0-2	5	-	4	PC	Smart Infrastructure Engineering	3-0-0	3	-	3
	Environmental Engineering	3-0-2	5	-	4						
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives			-	12
PE/ EE	Programme/Engineering Science Electives			-	10	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total			**-**	**23**		**Total**			**-**	**18**
										Total of graded and Non-Graded Credit	**9**	**160**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**15**	**175**

5) B.Tech. in Computer Science and Engineering + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						**VI Semester**				
PC	Maths for Computing	3-1-0		-	4	PC	Software Engineering	3-0-2		-	4
PC	Operating Systems	3-0-2			4	PC	Human-Machine Interaction	0-0-4			2
PC	Design and Analysis of Algorithms	3-1-0			4	PC	Digital Design	2-0-2			3
PC	Computer Architecture	3-0-0			3	PC	Computer Networks	3-0-3			4.5
PC	Principles of Programming Languages	3-0-0			3	PC	Digital Systems Lab	0-0-3			1.5
PE/ EE	Program/ES elective in lieu of DSA	3-0-0			3	PC	Cyber Security	3-0-2			4
IH	Humanities II	3-0-0	3	-	3	PE/ EE	Programme/Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			**1**	**24**		**Total**	**15-2-8**		**1**	**22**
	VII Semester						**VIII Semester**				
PC	Database Systems	3-0-2		-	4	PC	CS Core Bouquet 1	3-0-0		-	3
PC	CS Core Bouquet 2	3-0-0			3						
PP	B. Tech. Project	3-0-0	3	-	3	PE/ EE	Programme/Engineering Science Electives			-	14
PE/ EE	Programme/Engineering Science Electives			-	12						
IS	Environmental Science	2-0-0	2	-	2	IH	Humanities III	3-0-0	3	-	3
	Total			**-**	**21**		**Total**			**-**	**20**
										9	**160**
										6	**-**
										15	**176**

6) B.Tech. in Electrical Engineering + Engineering Science

Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
	V Semester						**VI Semester**				
PC	***Programme Compulsory of EE***			-	***18***	PC	***Programme Compulsory of EE***			-	***21***
	Analog Circuits	3-0-3	6		4.5		Digital Design	2-0-2	4		3
	Circuit Theory	3-0-0	3		3		Engineering Electromagnetics	3-0-0	3		3
	Control Systems	3-0-3	6		4.5		Physical Electronics	3-0-0	3		3
	Communication Systems	3-0-0	3		3		Data Communication Networks	3-0-0	3		3
	Computer Architecture	3-0-0	3		3		Digital Signal Processing	3-0-0	3		3
							Embedded Systems	3-0-0	3		3
							Communication Systems Lab	0-0-3	3		1.5
							Digital Systems Lab	0-0-3	3		1.5
IH	Humanities II	3-0-0	3	-	3	PE/ ES	Programme/Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
Total		**18-1-6**		**1**	**21**	**Total**		**20-1-8**		**1**	**24**
	VII Semester						**VIII Semester**				
PC	***Programme Compulsory of EE***			-	***3***	PC	***Programme Compulsory of EE***			-	***6***
	Electrical Machines	3-0-0	3		3		Power Engineering	3-0-3	6		4.5

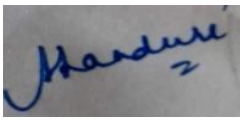
							Electric Machines Lab	0-0-3	3		1.5
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives			-	12
PE/ EE	Programme/Engineering Science Electives			-	13	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
Total				**-**	**21**	**Total**				**-**	**21**
Total of graded and Non-Graded Credit										**9**	**160**
Non-Graded Design Credits										**6**	**-**
Grand Total										**15**	**175**

7) B.Tech. in Mechanical Engineering + Engineering Science

Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
	V Semester						**VI Semester**				
	Manufacturing Processes and Systems lab	0-0-2	2		1		Design of Machine elements	3-0-2	5		4
	Heat Transfer	3-0-0	3		3		Fluid Mechanics-Lab	0-0-2	2		1
	Kinematics and Dynamics of Machines	3-0-2	5		4		Machining and Automated Manufacturing Lab	0-0-2	2		1
	Machining Science and Metrology	3-0-0	3		3		Smart Manufacturing	2-0-2	4		3
	Mechanics of solids	3-1-0	4		4		Thermofluids Engineering	3-1-0	4		4
	Fluid Mechanics	3-1-0	4		4		Control of Mechanical Systems	3-0-4	7		5
	Heat Transfer Lab	0-0-2	2		1		Manufacturing Process	3-0-0	3		3
IH	Humanities II	3-0-0	3	-	3	PE/ EE	Programme/Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total	**18-3-6**	**26**	**1**	**23**		**Total**	**17-2-12**		**1**	**24**
	VII Semester						**VIII Semester**				
PC	Production & Operations Management	3-0-0			3						
	Digital Fabrication	3-0-0			3						
	Thermofluids Lab	0-0-2			1						
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives		-	-	15
PE/ EE	Programme/Engineering Science Electives	-	-	-	10	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total	**-**	**-**	**-**	**22**		**Total**			**-**	**18**
										Total of graded and Non-Graded Credit	**9**	**160**
										Non-Graded Design Credits	**6**	**-**
										Grand Total	**15**	**175**

8) B.Tech. in Materials Engineering + Engineering Science

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	V Semester						VI Semester				
PC	Mechanics of Solids	3-1-0	4	-	4	PC	Materials for Energy Conversion and Storage	3-0-0	3		3
PC	Electronic Materials	3-0-0	3		3	PC	Computational Modelling of Materials	3-0-0	3		3
PC	Phase Transformations	3-0-0	3		3	PC	Materials Processing and Manufacturing	3-0-0	3		3
PC	Materials Characterization	3-0-0	3		3	PC	Materials at Equilibrium	3-1-0	4		4
PC	Mechanical Behavior of Materials	3-0-0	3		3	PC	Iron and Steel Making	3-0-0	3		3
PC	Electronic Materials Lab	0-0-3	3		1.5	PC	Smart Materials	3-0-0	3		3
PC	Materials Characterization Lab	0-0-2	2		1	PC	Computational Modelling of Materials Lab	0-0-2	2		1
PC	Physical Metallurgy Lab	0-0-3	3		1.5	PC	Materials Processing and Manufacturing Lab	0-0-3	3		1.5
IH	Humanities II	3-0-0	3	-	3	PE/ EE	Programme/Engineering Science Electives	3-0-0	3	-	3
NH	Professional Ethics I	0-1-0		1	-	NH	Professional Ethics II	0-1-0		1	-
	Total			1	23		Total	15-2-8		1	24.5
	VII Semester						VIII Semester				
PC	Energy Materials Lab	0-0-3	3		1.5	PC	Extraction of Nonferrous Metals	3-0-0	3		3
PP	B. Tech. Project	0-0-3	3	-	3	PE/ EE	Programme /Engineering Science Electives			-	14
PE/ EE	Programme/Engineering Science Electives			-	13	IH	Humanities III	3-0-0	3	-	3
IS	Environmental Science	2-0-0	2	-	2						
	Total			-	19.5		Total		20	-	20
										Total of graded and Non-Graded Credit	9	160
										Non-Graded Design Credits	6	-
										Grand Total	15	175



(Ashok Kumar Khanduri)
Joint Registrar (Academics)

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